Example 7.13 The slant height of a frustum of a cone is 5 cm and the radii of its ends are 4 cm and 1 cm. Find its curved surface area.

Solution Let l, R and r be the slant height, top radius and bottom radius of the frustum.

Given that, $l = 5$ cm, $R = 4$ cm, $r = 1$ cm

Now, C.S.A. of the frustum $= \pi(R + r)l$ sq. units

$$= \frac{22}{7} \times (4 + 1) \times 5$$

$$= \frac{550}{7}$$

Therefore, C.S.A. $= 78.57$ cm^2

Thinking Corner

1. Give two real life examples for a frustum of a cone.
2. Can a hemisphere be considered as a frustum of a sphere.

Example 7.14 An industrial metallic bucket is in the shape of the frustum of a right circular cone whose top and bottom diameters are 10 m and 4 m and whose height is 4 m. Find the curved and total surface area of the bucket.

Solution Let h, l, R and r be the height, slant height, top radius and bottom radius of the frustum.

Given that, diameter of the top $= 10$ m; radius of the top $R = 5$ m.

diameter of the bottom $= 4$ m; radius of the bottom $r = 2$ m, height $h = 4$ m

Now, $l = \sqrt{h^2 + (R - r)^2}$

$$= \sqrt{4^2 + (5 - 2)^2}$$

$$l = \sqrt{16 + 9} = \sqrt{25} = 5\text{m}$$

C.S.A. $= \pi(R + r)l$ sq. units

$$= \frac{22}{7}(5 + 2) \times 5 = 110\, m^2$$

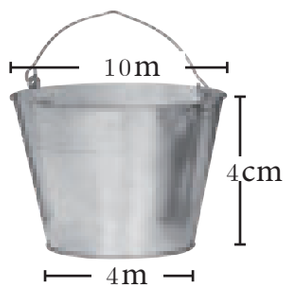


Fig. 7.24

T.S.A. $= \pi(R + r)l + \pi R^2 + \pi r^2$ sq. units

$$= \frac{22}{7}\left[(5 + 2)5 + 25 + 4\right] = \frac{1408}{7} = 201.14$$

Therefore, C.S.A. $= 110$ m^2 and T.S.A. $= 201.14$ m^2

Progress Check

1. The portion of a right circular cone intersected between two parallel planes is _____.
2. How many frustums can a right circular cone have?

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